Img2Img Regeneration Attack on Neural Image Watermarks: From VideoSeal to SynthID

Diffusion Watermarking Project

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Abstract

We evaluate the robustness of neural image watermarks against a Stable Diffusion img2img regeneration attack. **No training or fine-tuning is involved:** the attack uses an unmodified, off-the-shelf Stable Diffusion 1.5 checkpoint — a single forward pass through the img2img pipeline is sufficient. A coarse grid search against Meta's **VideoSeal** (256-bit, open-source) shows the watermark is **removed in 100% of trials** (30/30) at all tested strengths (0.3–0.7), establishing that even moderate img2img regeneration destroys the watermark signal. We then scale the evaluation to Google's **SynthID** (136-bit, closed-source): a pilot test verified via Google's detection portal showed removal at strength = 0.2; we now have a **50-image SynthID corpus** generated via the Gemini API with automated detection via the reverse-SynthID [6] open-source detector. Preliminary results on 2 images show that while the watermark signal degrades monotonically with attack strength (Fourier-domain correlation drops $\sim$3$\times$ from original to $s = 0.3$), the detector's binary decision remains positive, suggesting SynthID may be more robust than VideoSeal or that higher attack strengths are needed. A full 50-image sweep at six fine-grained strengths (0.05–0.30) is in progress. The attack requires no knowledge of the watermark's architecture or parameters.

1 Introduction

Image watermarking is increasingly deployed to identify AI-generated content. Google's SynthID [2] embeds a 136-bit neural watermark at generation time, but its detection model is closed-source. To study watermark robustness against diffusion-based attacks without access to SynthID, we use open-source proxies.

VideoSeal [1] (Meta FAIR, 2024) is the closest open-source analogue to SynthID: both use learned neural encoder/decoders operating post-hoc on images, both embed multi-bit payloads (256 bits for VideoSeal, 136 for SynthID), and both are designed for imperceptible embedding.

Our attack is deliberately minimal: pass the watermarked image through an **unmodified, pretrained** Stable Diffusion img2img pipeline. No training, fine-tuning, or adversarial optimisation is performed. The diffusion model is used purely for inference — a single img2img call that adds noise (destroying the watermark's learned signal) then denoises (reconstructing a visually similar image). The attacker needs no knowledge of the watermark's architecture, weights, or detection threshold.

We use a coarse grid search over attack strength and guidance scale to characterise the quality–robustness tradeoff, not to "tune" the attack: VideoSeal fails at every tested setting. We then scale to SynthID with a 50-image Gemini-generated corpus and automated detection via reverse-SynthID. The key open question is *what is the minimum attack strength that reliably removes each watermark* — answering this for SynthID requires the fine-grained sweep now in progress.

2 Experimental Setup

2.1 Watermark Configuration

We use the `videoseal_1.0` model (checkpoint `y_256b_img.pth`, 228 MB). The embedder operates in YUV colour space with a UNet-based architecture; the extractor is a ConvNeXt-Tiny producing 257 outputs (1 presence indicator + 256 message bit logits).

- **Message:** Deterministic 256-bit pseudorandom sequence (seed = 42), with 126 ones and 130 zeros.
- **Embedding strength:** `scaling_w` = 0.3 (50% above the default 0.2).
- **Detection threshold:** $p < 0.01$ via one-sided binomial test on bit matches.

2.2 Test Images

Five images from DiffusionDB [3] (Stable Diffusion–generated, 512×512 PNG), the target domain for AI-generated content watermarking.

2.3 Attack Model

Stable Diffusion v1.5 [4] with DDIM scheduler, loaded in `float32` on Apple MPS with attention slicing enabled. The attack wraps the pipeline as `StableDiffusionImg2ImgPipeline` and applies regeneration at varying strength.

Note: The original plan used SD 2.1 (768×768), but Stability AI removed it from HuggingFace (EU AI Act compliance) and it was prohibitively slow on MPS (∼17 min/step). SD 1.5 at 512×512 runs at ∼2 steps/s — a ∼1000× speedup. Since the attack model is purely a "weapon" (independent of the watermark method), any capable diffusion model suffices.

2.4 Grid Search Parameters

Table 1: Attack hyperparameter grid.

Parameter	Values	Description
Attack strength	$\{0.3, 0.5, 0.7\}$	Fraction of diffusion steps applied
Guidance scale	$\{1.0, 7.5\}$	Classifier-free guidance weight
Inference steps	50	Total DDIM steps
Seed	42	Random seed for reproducibility

Total: 6 parameter combinations × 5 images = 30 runs.

3 Baseline: Pre-Attack Performance

Before attacking, we verify that the watermark embeds and detects correctly at `scaling_w` = 0.3.

The watermark is effectively invisible (SSIM > 0.99, PSNR > 40 dB) and detection is perfect with a p-value gap of ∼76 orders of magnitude between watermarked and clean images.

4 Attack Results

4.1 Full Results

Table 3 shows all 30 runs. Every single configuration successfully removed the watermark (detected = False).

Table 2: Pre-attack watermark performance (`scaling_w` = 0.3).

	SSIM	PSNR (dB)	Bit Acc.	p-value	Detected
Watermarked images (true positives)					
Average (5 images)	0.9907	41.6	100.0%	8.6×10^{-78}	5/5
Clean images (false positives)					
DiffusionDB (5 images)	—	—	51.8%	~ 0.3	0/5
DIV2K (5 images)	—	—	50.8%	~ 0.5	0/5

Table 3: Full attack sweep results. All 30 runs: watermark **not detected**.

Atk str.	CFG	Image	SSIM	PSNR	p-value	Bit Acc.	Det.
0.3	1.0	0000	0.363	18.3	1.2×10^{-1}	53.9%	No
		0001	0.538	18.3	1.7×10^{-1}	53.1%	No
		0002	0.552	19.2	2.9×10^{-1}	52.0%	No
		0003	0.686	22.8	9.5×10^{-2}	54.3%	No
		0004	0.559	23.5	1.7×10^{-1}	53.1%	No
0.5	1.0	0000	0.250	16.4	1.2×10^{-1}	53.9%	No
		0001	0.390	15.2	6.7×10^{-1}	48.8%	No
		0002	0.467	17.4	8.6×10^{-1}	46.9%	No
		0003	0.604	20.4	3.3×10^{-1}	51.6%	No
		0004	0.454	21.7	7.1×10^{-1}	48.4%	No
0.7	1.0	0000	0.186	15.2	9.1×10^{-1}	46.1%	No
		0001	0.254	12.3	6.7×10^{-1}	48.8%	No
		0002	0.365	15.4	7.5×10^{-1}	48.1%	No
		0003	0.521	17.6	7.5×10^{-2}	54.7%	No
		0004	0.360	19.4	4.8×10^{-1}	50.4%	No

Results for CFG = 7.5 omitted from this table: they are **identical** to CFG = 1.0 in every metric (see Section 4.3).

4.2 Aggregated by Attack Strength

Table 4 shows the mean metrics across all 5 images for each attack strength (pooling over both guidance scales, which are identical).

Table 4: Mean attack metrics by strength (averaged over 5 images).

Atk str.	SSIM	PSNR (dB)	Bit Acc.	Detected	Time (s/img)
0.3	0.540	20.4	53.1%	0/5	10.1
0.5	0.433	18.2	49.9%	0/5	18.0
0.7	0.337	16.0	49.6%	0/5	28.3
Chance level	—	—	*50.0%*	—	—

4.3 Guidance Scale Has No Effect

A surprising finding: classifier-free guidance scale (CFG) has **zero effect** on attack efficacy. For every (image, strength) pair, the SSIM, PSNR, p-value, and bit accuracy are identical between CFG = 1.0 and CFG = 7.5.

This makes physical sense: the img2img attack uses an empty prompt. Classifier-free guidance interpolates between unconditional and conditional predictions, but with an empty prompt

both predictions are identical, so the guidance scale is a no-op.

Implication: Future sweeps can fix CFG = 1.0 and focus the parameter budget on finer strength granularity.

5 Analysis

5.1 Watermark Removal is Trivially Easy

The watermark was removed in **100% of trials** (30/30) at all tested attack strengths (0.3–0.7). Post-attack bit accuracy is indistinguishable from chance (50%), with p-values of $\mathcal{O}(10^{-1})$ compared to $\mathcal{O}(10^{-78})$ pre-attack — a collapse of $\sim$77 orders of magnitude.

5.2 Optimal Operating Point

The Pareto frontier is straightforward. Attack strength = 0.3 is the clear winner:

- **Quality:** SSIM ≈ 0.54, PSNR ≈ 20 dB (moderate degradation but images remain recognisable).
- **Efficacy:** Bit accuracy drops to 53% (only 3 percentage points above chance, well within the noise floor).
- **Speed:** $\sim$10 s per image on Apple MPS.

Stronger attacks (0.5, 0.7) destroy more image quality without meaningfully improving watermark removal — the watermark is already gone at 0.3.

5.3 Why the Attack Works

The img2img regeneration attack succeeds because:

1. **Information destruction:** The forward diffusion process (adding noise proportional to the strength parameter) directly destroys the learned watermark signal embedded in pixel-space activations.
2. **Signal reconstruction:** The reverse diffusion process (denoising) reconstructs a plausible image from the noised latent, but the reconstructed image carries the *diffusion model's* implicit prior rather than the watermark encoder's signal.
3. **Dimensional mismatch:** VideoSeal embeds information in a 256-dimensional binary code. The diffusion process operates in a much higher-dimensional latent space ($4 \times 64 \times 64 = 16{,}384$ dimensions for SD 1.5), and even 30% noise injection perturbs enough dimensions to destroy the watermark's statistical signature.

6 Transfer to SynthID

Having tuned attack parameters on VideoSeal, we apply the same img2img regeneration attack to **SynthID-watermarked images** generated by Google Gemini (Imagen 3). SynthID is Google's closed-source 136-bit neural watermark, deployed in production across Gemini-generated images.

6.1 Pilot: Single Image via Google's Portal

The initial test used a single image ("old man in a boat", 1408×768) generated via the Gemini API, with detection verified through Google's public SynthID detection portal (binary verdict only).

Because SD 1.5 operates natively at 512×512, we use a **tiled attack**: the image is split into overlapping 512×512 patches, each attacked independently, then reassembled with linear feathering. Prior to attack, the visible Gemini badge was removed via OpenCV inpainting and non-essential PNG chunks were stripped.

Table 5: Pilot SynthID attack ($n = 1$, 1408×768, tiled at 512×512). Detection via Google's portal.

Atk str.	SSIM	PSNR (dB)	SynthID Detected
0.2	0.684	22.1	**No**
0.3	0.617	20.7	**No**
0.5	0.472	18.3	**No**
Original	*1.000*	∞	*Yes*

All three strengths removed the watermark according to Google's detector. However, this is a single data point and the portal returns only a binary verdict with no confidence score.

6.2 Corpus Evaluation: 50 Images via reverse-SynthID

To scale beyond $n = 1$, we generated a **50-image SynthID corpus** via the Gemini API (diverse prompts: landscapes, portraits, abstract art, food, sci-fi, etc.; all 1024×1024 PNG). For automated detection we use **reverse-SynthID** [6], an open-source detector that analyses Fourier-domain carrier frequencies embedded by SynthID.

6.2.1 Detection Methodology

Reverse-SynthID extracts a noise residual via multi-denoiser fusion (wavelet, bilateral, non-local means, Wiener) and checks for SynthID's known carrier frequencies in the Fourier domain. Detection combines four signals:

1. **Correlation** with a reference noise pattern (built from a codebook of watermarked images).
2. **Phase match** at known carrier frequencies (e.g. $(14, 14)$, $(126, 14)$, $(128, 128)$).
3. **Noise structure ratio** (standard deviation / mean absolute value of the extracted noise).
4. **Multi-scale consistency** across 256, 512, and 1024 px resolutions.

We built a **corpus-specific codebook** from all 50 watermarked images (correlation mean $= 0.001$, detection threshold $= -0.007$).

Detector limitations: The binary `is_watermarked` decision has a high false positive rate — even random noise passes the loose threshold. The *continuous metrics* (correlation, phase match) are far more informative. We report both but draw conclusions from the continuous signals.

6.2.2 Tiled Attack

Each 1024×1024 corpus image is split into a 2×2 grid of non-overlapping 512×512 tiles (matching SD 1.5's native resolution). Each tile is attacked independently via img2img regeneration, then reassembled. This yields 4 tile attacks per image.

Attack parameters: guidance scale = 1.0 (irrelevant with empty prompt, per Section 4.3), 50 DDIM steps, seed = 42.

6.2.3 Preliminary Results ($n = 2$)

Table 6 shows results from a preliminary 2-image test at strengths 0.10 and 0.30.

Table 6: SynthID corpus attack (preliminary, $n = 2$). Continuous metrics from reverse-SynthID detector.

Image	Atk str.	SSIM	PSNR	Correlation	Phase Match	Confidence
0000	*Original*	1.000	∞	0.0645	0.677	1.000
	0.10	0.792	26.7	0.0292	0.581	1.000
	0.30	0.732	24.5	0.0182	0.565	1.000
0001	*Original*	1.000	∞	0.0928	0.603	1.000
	0.10	0.604	25.2	0.0488	0.541	1.000
	0.30	0.506	22.2	0.0309	0.569	1.000

Key observations:

- **Correlation degrades monotonically:** From 0.065 / 0.093 (original) to 0.018 / 0.031 at $s = 0.3$ — a 2–3× reduction. This confirms the attack is degrading the watermark signal.
- **Phase match degrades:** From 0.68 / 0.60 to 0.57 / 0.57 at $s = 0.3$, approaching the detection threshold of 0.45.
- **Binary detector still positive:** At $s = 0.3$, the detector's binary decision remains "detected" (confidence = 1.0). However, this is partly due to the detector's loose thresholds — even random noise produces confidence ≈ 0.79.
- **SynthID appears more robust than VideoSeal:** VideoSeal's bit accuracy dropped to chance at $s = 0.3$; SynthID's continuous metrics remain well above noise floor. This suggests either (a) SynthID is architecturally more resilient, or (b) the reverse-SynthID detector is less sensitive than VideoSeal's extractor.
- **Higher strengths needed:** The full sweep ($s \in \{0.05, \ldots, 0.30\}$ on all 50 images) will clarify whether SynthID breaks at these strengths or requires $s > 0.3$.

6.3 Discussion

The combined evidence from the pilot and corpus evaluations is nuanced:

1. **Google's portal says removed at** $s = 0.2$ (Section 6.1), but Google's detector is a black box — we cannot distinguish a hard threshold flip from a marginal decision.
2. **reverse-SynthID's continuous metrics show clear degradation** but not yet removal at $s = 0.3$. The correlation signal remains ∼20× above random noise.

3. **The two detectors may have different sensitivities.** Google's production detector (trained on SynthID's exact architecture) is likely more accurate than the open-source reverse-engineering effort. If Google's detector returns "not detected" at $s = 0.2$ while reverse-SynthID still sees signal at $s = 0.3$, the watermark may be functionally broken even when residual Fourier artifacts remain.

The resolution: the full 50-image sweep will produce a degradation curve that clarifies whether SynthID has a sharp phase transition (like VideoSeal) or a gradual decline.

7 Limitations

- **VideoSeal sample size:** Tested on 5 DiffusionDB images only. Full DiffusionDB (1,000) and DIV2K (900) sweeps are prepared but not yet run.
- **SynthID sweep in progress:** Corpus of 50 images is ready; full 6-strength sweep (300 runs) not yet complete. Preliminary results cover only 2 images $\times$ 2 strengths.
- **Single watermark strength:** Only `scaling_w` $= 0.3$ was attacked for VideoSeal. Higher embedding strengths may be more robust.
- **SD 1.5 only:** Different attack models (SDXL, SD 3.x, Flux) may have different attack profiles. However, since even the "weakest" model trivially removes VideoSeal, stronger models are unlikely to change the qualitative conclusion.
- **No adaptive defence:** Neither VideoSeal nor SynthID was fine-tuned to resist regeneration attacks. An adversarially trained watermark may fare better.
- **Quality metrics only:** SSIM and PSNR measure pixel-level similarity but do not capture perceptual or semantic quality. FID or LPIPS would be more informative.
- **reverse-SynthID detector accuracy:** The open-source detector has a high false positive rate (random noise achieves confidence ≈ 0.79, phase match ≈ 0.47). Its binary decision is unreliable; we rely on continuous metrics (correlation, phase match) for quantitative analysis. The discrepancy between Google's portal (which reports removal at $s = 0.2$) and reverse-SynthID (which still detects at $s = 0.3$) is unresolved.

8 Next Steps

1. **Complete SynthID sweep** (*in progress*): Run the full 50-image $\times$ 6-strength sweep (300 attacks). Infrastructure and corpus are ready; execution takes $\sim$5 hours on MPS.
2. **Extend SynthID strength range:** If the watermark survives at $s = 0.30$, extend to $\{0.35, 0.40, 0.50, 0.60, 0.70\}$ to find the breaking point.
3. **Calibrate reverse-SynthID thresholds:** The current detector has a $\sim$80% false positive rate on random noise. Build a calibration set (50 watermarked + 50 clean images) to find a correlation/phase-match threshold with $< 5\%$ FPR.
4. **Scale VideoSeal to full datasets:** Run on all 1,000 DiffusionDB and 900 DIV2K images with finer strengths.
5. **Multiple watermark methods:** Compare VideoSeal, Stable Signature, and Tree-Ring under the same attack.
6. **Perceptual metrics:** Add LPIPS and FID to the quality evaluation.

7. **Tiling analysis:** Study whether tile boundaries create detectable artefacts that a forensic detector could exploit.

9 Reproduction

9.1 VideoSeal Sweep

```
# Embed watermarks (5 images, scaling_w=0.3)
python scripts/embed.py --method videoseal \
  --input-dir "datasets/diffusiondb" \
  --num-images 5 --strengths 0.3

# Run attack sweep (6 combos x 5 images)
python scripts/attack_sweep.py --max-images 5

# Results: $DW_RESULTS_DIR/attack_videoseal_diffusiondb_s0.3.csv
```

9.2 SynthID Sweep

```
# Clone reverse-SynthID detector
git clone https://github.com/aloshdenny/reverse-SynthID
pip install PyWavelets scikit-learn tqdm opencv-python

# Build codebook from 50 SynthID corpus images
python reverse-SynthID/src/extraction/robust_extractor.py \
  extract test-synthID/corpus/ \
  --output reverse-SynthID/artifacts/codebook/corpus_codebook.pkl

# Quick test (2 images x 2 strengths)
python scripts/attack_synthid_sweep.py \
  --max-images 2 --strengths 0.10 0.30

# Full sweep (50 images x 6 strengths = 300 attacks)
python scripts/attack_synthid_sweep.py

# Results: test-synthID/results/synthid_sweep.csv
# Attacked images: test-synthID/attacked/s{strength}/
```

Hardware: Apple M-series (MPS), ∼14 minutes for 30 VideoSeal runs; ∼30 s per SynthID attack (4 tiles × 50 DDIM steps), ∼5 hours estimated for the full 300-run sweep. Software: PyTorch 2.1+, diffusers, VideoSeal, reverse-SynthID, OpenCV.

References

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